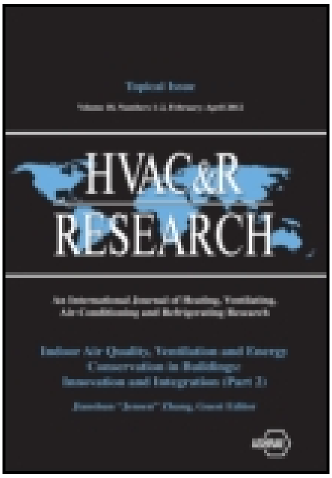

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Dynamic modeling for vapor compression systems—Part I: Literature review

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Review Article
Dynamic modeling for vapor compression systems—Part I: Literature review

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This two-part article provides an introduction to dynamic modeling for vapor compression systems. Part I provides a detailed review of current literature in this area. Both physics- and data-based approaches are discussed with their associated advantages and limitations. Physics-based modeling paradigms include (1) lumped parameter approaches that qualitatively capture gross pressure and cooling transients, (2) moving boundary approaches that seek to model the dynamic variations in phase transition points, and (3) finite-control volume approaches that use discretized models that include temperature and parameter gradients in an effort to achieve greater accuracy. These models are based on first principles, but yet require time-consuming tuning and validation with experimental data. Data-based approaches offer faster model generation but are specific to the system and conditions from which the data originated.

Introduction

Vapor compression cycles are used widely for air conditioning and refrigeration. In their simplest form, a vapor compression system (VCS) consists of two heat exchangers, an expansion valve, and a compressor (Figure 1). The ideal VCS consists of four processes: (1) isentropic compression, (2) isobaric heat rejection and condensation, (3) isenthalpic expansion, and (4) isobaric heat absorption and evaporation (Figure 2). The dynamics are driven by fluctuating pressures of the two-phase fluid in the heat exchangers, and fluctuating refrigerant mass flow rates through the compressor and expansion valve. The complexity of the two-phase fluid dynamics adds a richness to VCS behavior that can be challenging to model.

This two-part article seeks to provide a tutorial regarding dynamic modeling and simulation of VCSs. In part I, a comprehensive review of related literature is presented; in part II, an overview of simulation models is given and issues of integration, validation, and computation will be addressed.

Dynamic models are often classified using such terms as white box, gray box, or black box. The term white-box models refer to physics-based models that are described using physical laws, such as conservation equations. These models also appear in the literature as "mechanistic" models or "first-principles" models. At the other extreme, black-box models refer to empirical or data-driven models, where transient experimental data is used to identify a dynamic model. This process is also known as system identification or time-series analysis, and

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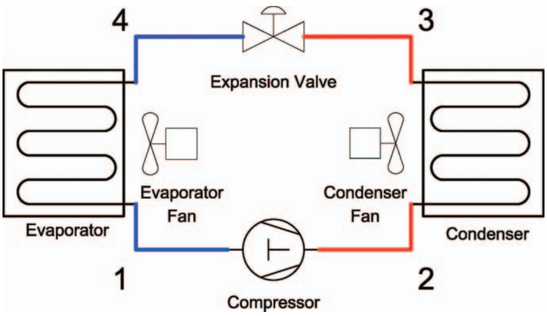

Figure 1. Subcritical vapor compression cycle: system diagram (color figure available online).

it can be used to construct models in the time or frequency domain. The bulk of modeling efforts for VCSs are most appropriately termed gray-box, as they are largely based on the governing physics but include semi-empirical terms, such as efficiency maps, heat transfer correlations, etc., that are created using data from experimental tests.

As will be discussed in subsequent sections, the advantage of a data-driven modeling approach is expediency, as a black-box model can generally be identified quickly from data. This model, however, is typically only valid at the tested condition and for the specific system tested. Although physics-based models are more time-intensive to create, tune, and validate, they are more robust, as changes in system parameters or operating conditions can be incorporated immediately into the governing equations. In this article both modeling approaches will be discussed in the specific context of VCSs.

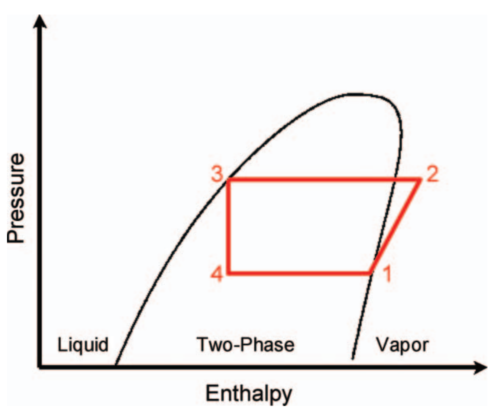

Figure 2. Subcritical vapor compression cycle: pressure-enthalpy diagram (color figure available online).

Physics-based dynamic models

In this section, physics-based modeling approaches for each component of a typical VCS are presented, and published approaches will be classified and discussed. A VCS model is constructed by appropriately integrating component models (see Figure 1 in Part II of this article). The primary focus is on two-phase heat exchanger dynamics that determine the evolution of system pressures with inlet/outlet mass flow rates as model inputs. Mass flow devices, such as compressors and valves are also discussed briefly, as these models are significantly simpler and utilize inlet and outlet pressures to determine mass flow rates. Outlet fluid enthalpy is determined by each component model and supplied as an input to subsequent component models. Receiver models are addressed separately. After individual component models are reviewed, fundamental theoretical and practical challenges associated with VCS model simulation and prediction are discussed.

Compressors

Many residential, commercial, and industrial VCSs are driven by positive displacement compressors. Specific studies of the mechanical dynamics of these compressors are available in the literature, for example, reciprocating (Longo and Gasparella 2003), scroll (Chen et al. 2004), vane (Al-Hawaj 2009), linear (Kim et al. 2009), and electrostatic (Sathe et al. 2010). However, since the dynamics of these compressors typically evolve on much faster time scales than the heat exchanger dynamics, these dynamics are typically not included when a compressor model is integrated into a VCS model. For example, all publications that employ the low-order heat exchanger models use static relationships that dictate the compressor mass flow rate and the outlet fluid enthalpy (Table 1). These relationships are defined by a volumetric efficiency for determining refrigerant mass flow rate and an adiabatic efficiency for determining compressor work (see Part II of this article for representative equations). When more high-order heat exchanger models are employed, static compressor models are generally still employed, but there are some publications that include dynamic models for the mechanical dynamics (with varying degrees of detail) and the thermal capacitance of the compressor shell (Table 2). The other exception is centrifugal compressors, as they exhibit unique surge-and-stall dynamics. These were examined at length by Greitzer and Moore (1986) and Moore and Greitzer (1986) and subsequent works.

Expansion valves

Expansion valves are a primary means of metering refrigerant flow in a VCS. Orifice and capillary valves (OEV) have a fixed area, and they are designed to meter the proper flow of refrigerant at a particular condition. More complex valves utilize mechanical or electronic means to regulate refrigerant flow based on pressure or temperature. Thermostatic expansion valves (TEVs) utilize a sensing bulb filled with a two-phase refrigerant placed at the evaporator outlet to measure superheat temperature. The bulb pressure acts on a diaphragm inside the bulb to increase/decrease the flow area accordingly. Expansion valves that alter the flow area to regulate pressure, rather than superheat, are termed pressure regulation valves, or automatic expansion valves (AEVs). These work on the same principle as the TEV, but instead of a sensing bulb, they use the evaporator pressure directly. Electronic expansion valves (EEVs) modify the flow area using an externally controlled mechanism, typically a stepper motor.

The published literature on expansion valves is heavily concentrated on TEV and EEV dynamics, although some work presents models specific to fixed orifice valves, e.g., capillary tubes (Murphy and Goldschmid 1985; García-Valladares 2007a, 2007b). This emphasis is due to the propensity of TEV-controlled evaporators to experience valve hunting, a condition is characterized by oscillations in the length of two-phase flow in the evaporator and, thus, oscillations in the amount of superheated vapor at the evaporator exit. This behavior was first qualitatively documented in Zahn (1964). Wedekind and Stoecker (1966) published data demonstrating this phenomenon along with a dynamic model of a TEV-driven evaporator, where the evaporator is modeled with two first-order dynamics with a time delay and an additional dynamic for the valve (Stoecker 1966). The associated parameters were identified for various operating conditions using experimental data. Najork (1973) gave recommendations for parameter adjustments to minimize hunting, based on a simplified physical model. Broersen and van der Jagt (1980) used a mean void fraction model of the evaporator to show that valve hunting is caused by the interaction between the TEV and evaporator dynamics, and three years later, Broersen and ten Napel (1983) presented an identified model of the TEV's sensing bulb dynamics. Higuchi and Hayano (1982) presented a detailed linear dynamic model of the TEV and validated the associated frequency reponse. Yasuda et al. (1983) presented a validation of a fourth-order TEV model and a VCS that qualitatively captures the hunting behavior, if not quantitatively perfect. Gruhle and Isermann (1985) also used a discretized model of the evaporator to contrast the TEV with the proportional-integral (PI) controlled valve. Although authors continued to report TEV models with increasing levels of physical detail (James and James 1987; Ibrahim 2001), none of the cited publications provide any experimental validation; the only exceptions are those noted (e.g., Higuchi and Hayano 1982; Yasuda et al. 1983).

Recent efforts have seen a return to simpler first-order lumped models for the TEV dynamics. Mulay et al. (2005) analytically derived an approximate time constant associated with the sensing bulb of a TEV and then explored the impact of bulb location and time constant on superheat control. Mithraratne et al. (2000) also presents a physics-based model of the TEV-controlled evaporator using a discretized evaporator model and a first-order dynamic TEV model. While the validation presented shows some improvement compared to the moving boundary (MB) evaporator models, subsequent validation in Mithraratne and Wijeysundera (2002) exhibited significant discrepancies. Recent efforts have sought to develop parameter estimation algorithms for TEV models to facilitate fast model development and improve validation results (Hariharan and Rasmussen 2010).

Although comparatively fewer publications have examined EEV-controlled evaporators, these are appearing with increasing frequency, with a particular focus on control algorithm design. For example, in Outtagarts et al. (1997), a simplified identified model was used for the evaporator, and then proportional-integral-derivative (PID) and optimal control algorithms compared. Similarly, in Finn and Doyle (2000), a PID controller based on a simplified identified model for the evaporator was designed, and PID control was compared to the TEV. While the EEV offers the capability for more sophisticated control strategies, most exhibit highly nonlinear behavior, leading many researchers to suggest that scheduling of controller gains is necessary to effective operation (Outtagarts et al. 1997; Finn and Doyle 2000), while other work suggests

Table 1. Summary of MB models[a]

		Heat exchanger		Auxiliary components			
Citation	System	Paradigm	Interface wall temperature	Compressor	Valve	Receiver	Transient validation
Wedekind et al. (1978) Beck and Wedekind (1981)	Two-phase heat exchanger	MB evaporator/condenser	NA	None	None	None	Yes
Bendapudi (2005) Bendapudi et al. (2008)	Centrifugal chiller	SMB evaporator/condenser	Two-phase wall temperature	Static with delay	TEV first order	None	Yes
Broersen and van der Jagt (1980)	TEV-controlled evaporator	MB evaporator	Time varying	None	TEV third order	None	No
Eisenhower and Runolfsson (2009)	Transcritical vapor compression	MB evaporator	Lumped wall temperature	Static	None	None	Yes
Eldredge et al. (2008)	Vapor compression	EMB evaporator/condenser	Two-phase wall temperature	Static	EEV, static	Integrated receiver/Heat exchanger (HX)	Yes
Gordon et al. (1999)	Vapor compression	SMB evaporator	Lumped wall temperature	Static	EEV, static	None	No
Grald and MacArthur (1992)	Two-phase heat exchangers	MB evaporator	Discretized wall temperature	None	None	None	Yes
Hariharan and Rasmussen (2010)	TEV-controlled evaporator	MB evaporator	Two-phase wall temperature	Static	TEV first order	None	Yes
Sheng et al. (1996) He et al. (1997) He et al. (1998)	Vapor compression	MB evaporator/condenser	Two-phase wall temperature	Static	EEV, static	None	Yes
Jensen and Tummescheit (2002)	Two-phase heat exchangers	MB evaporator	Switching wall temperature	Static	OEV, static	None	No
Kapadia and Wolgemuth (1984)	Rankine cycle	MB condenser	Two-phase wall temperature	None	None	None	No
Kumar et al. (2008)	Railway coach air conditioner	MB evaporator/condenser	Lumped wall temperature	None	None	None	Yes
Leducq et al. (2003)		MB evaporator/condenser	Lumped wall temperature	Static	EEV, static	None	Yes

(Continued on next page)

Table 1. Summary of MB models[a] *(Continued)*

		Heat exchanger		Auxiliary components			
Citation	System	Paradigm	Interface wall temperature	Compressor	Valve	Receiver	Transient validation
Lei and Zaheeruddin (2005)	Water chiller	MB evaporator/condenser	Neglected	Static	TEV, static	None	No
McKinley and Alleyne (2008) Li and Alleyne (2010)	Vapor compression	SMB evaporator/condenser	Switching wall temperature	Static	EEV, static	None	Yes
Pettit et al. (1998) Willatzen et al. (1998)	Refrigeration systems	SMB evaporator/condenser	Two-phase wall temperature	None	None	None	No
Rasmussen and Alleyne (2004) Rasmussen et al. (2005)	Transcritical vapor compression	MB evaporator	Two-phase wall temperature	Static	EEV, static	None	Yes
Schurt et al. (2009)	Refrigeration systems	MB evaporator/condenser	Neglected	Static	EEV, static	None	Yes
Zhang and Zhang (2006)	Evaporators	SMB evaporator	Weighted average	None	None	None	No

[a]In this table "transient validation" only indicates whether simulations were compared to experimental data, regardless of the number of variables compared, the quality of the agreement, or the appropriateness of the sampling rate used in collecting the data. Quality of validation can be a subjective judgment, and the cited publications exhibit significant differences in terms of the quality of model validation; the reader should take care to determine whether the published model is sufficiently validated for their intended purposes. This comment also applies to the publications listed in Table 2.

Table 2. Summary of FCV models

Citation	System	Heat exchanger: Paradigm	Heat exchanger: Momentum	Auxiliary components: Compressor	Auxiliary components: Valve	Auxiliary components: Receiver	Transient validation
Anand et al. (2004)	Automotive air conditioner	FCV evaporator/condenser	Included	Static	TEV, first order	Static	Yes
Beghi and Cecchinato (2009)	Fin-coil evaporator	FD evaporator	Included	Static	EEV, first order	None	Yes
Bendapudi and Braun (2002a) Bendapudi (2005) Bendapudi et al. (2008)	Centrifugal chiller	FCV evaporator/condenser	Neglected	Static with delay	TEV, first order	None	Yes
Chen and Lin (1991)	Small-scale refrigeration	FD evaporator/condenser	Included	Dynamic	Capillary	None	Yes
Cullimore and Hendricks (2001)	Vehicle air conditioner	FCV evaporator/condenser	Included	Static	OEV/TEV/ Capillary, static	Dynamic	No
García-Valladares et al. (2004)	Double pipe HX	FCV evaporator/condenser	Included	None	None	None	No
Gruhle and Isermann (1985)	Refrigerant evaporator	FCV evaporator	Included	Static	OEV, static	None	No
Hermes and Melo (2008)	Household refrigerator	FCV evaporator/condenser	Neglected	Dynamic	Capillary	None	Yes
Jia et al. (1995) Jia et al. (1999)	Evaporator	FD evaporator	Included	None	None	None	Yes
Kapadia et al. (2009)	Air conditioner	FD evaporator/condenser	Included	Static	Capillary	None	Yes
Koury et al. (2001)	Refrigeration system	FCV evaporator/condenser	Included	Static	TEV, static	None	Yes
MacArthur (1984) MacArthur and Grald (1989)	Heat pump	FCV evaporator/condenser	Neglected	Dynamic	TEV, first order	Dynamic	Yes

(Continued on next page)

Table 2. Summary of FCV models *(Continued)*

		Heat exchanger		Auxiliary components			
Citation	System	Paradigm	Momentum	Compressor	Valve	Receiver	Transient validation
Mithraratne et al. (2000) Mithraratne and Wijeysundera (2001) Mithraratne and Wijeysundera (2002)	Counter-flow evaporator	FD evaporator	Neglected	None	TEV, first order	None	Yes
Ndiaye and Bernier (2010)	Refrigerant-air fin-and-tube HX	FD evaporator/condenser	Included	None	None	None	Yes
Nyers and Stoyan (1994)	Heat pump evaporator	FD evaporator	Included	Static	EEV, static	None	No
Pfafferott and Schmitz (2000) Pfafferott and Schmitz (2004)	CO_2 Refrigeration	FCV evaporator	Included	Static	OEV, static	Static	No
Rigola et al. (2005)	Transcritical vapor compression	FCV evaporator	Included	Dynamic	OEV/Capillary, static	None	No
Sami et al. (1987) Sami and Comeau (1992)	Heat pump	FCV evaporator/condenser	Included	Static	TEV static/Capillary	Dynamic	Yes
Skaugen and Svensson (1998)	CO_2 heat pump	FCV evaporator	Neglected	Static	EEV, static	None	No
Tso et al. (2006a) Tso et al. (2006b)	Evaporator with frosting	FD evaporator	Included	None	None	None	No
Tummescheit and Eborn (1998) Tummescheit et al. (2000) Eborn (2001) Tummescheit (2002) Tummescheit et al. (2005)	Vapor compression	FCV evaporator/condenser	Included	Static	EEV, static	Dynamic	No
Wang et al. (2007)	Refrigeration system	FD evaporator/condenser	Neglected	Static	TEV, static	None	Yes
Wang and Touber (1991)	Plate fin air cooler	FD evaporator	Included	None	None	None	Yes
Zhang et al. (2009)	Residential refrigeration	FCV evaporator/condenser	Comparison	Static	EEV, static	Dynamic	No

alternative valve or controller configurations to eliminate nonlinear effects (Elliott et al. 2009; Elliott and Rasmussen 2010).

Receivers/accumulators

Receivers and accumulators are used in VCSs to store excess refrigerant and ensure that the fluid leaving the pressure vessel is in the appropriate fluid phase (e.g., refrigerant leaving the low-ide accumulator and entering the compressor is a vapor). Modeling of these components is straightforward, applying conservation of mass and energy to the refrigerant and conservation of energy to the receiver shell. While dynamic modeling of these components has not received as much attention as other VCS components, several different modeling approaches were developed and compared by Estrada-Flores et al. (2003).

Heat exchangers

Given that the thermal dynamics of a VCS are typically slower than the mechanical dynamics, the bulk of the model complexity generally resides in the heat exchangers. Previous literature reviews (Lebrun and Bourdouxhe 1998; Bendapudi and Braun 2002b) indicated that the bulk of the research efforts is focused on capturing two-phase flow dynamics in the heat exchangers while seeking a balance between simplicity and fidelity. For the purpose of this article, heat exchanger models will be classified into three categories. Lumped parameter models refer to models that apply lumped parameter assumptions to the entire heat exchanger or to particular fluid phases within the heat exchanger (i.e. individiual lumped models for superheated vapor, two-phase fluid, and subcooled liquid). In this review, the term "lumped parameter model" includes approaches that model the heat exchanger as a single-control volume or multiple (time-invariant) control volumes for each fluid phase. Most of the publications in the lumped parameter classification are early efforts, where computational simplicity is paramount to ensure feasible computation times, motivating modeling efforts with few dynamic equations and few (lumped) parameters. A variation of the lumped parameter approach is the MB model, where parameters are again lumped in regions defined by fluid phase, but the transition point between fluid phases is allowed to be a dynamic variable. Recent efforts in this area have expanded the paradigm to accommodate appearing/disappearing fluid phase regions. While these first two paradigms seek to create accurate models of relatively low dynamic order, finite-control volume (FCV) models arise from heavily discretizing a lumped parameter heat exchanger model into many zones or cells or by discretizing the governing partial differential equations (PDEs) that describe the heat exchanger with time and spatial variations. The number of works utilizing this more complex paradigm has increased in recent years in parallel with advances in computational capabilities. The obvious goal is increased accuracy through greater detail but at the expense of high dynamic order, which is more suited for simulation than insightful analysis or model-based control design.

Lumped parameter models

The extreme case of lumped parameter models that utilize approximate time constants will be addressed in a subsequent section in the context of data-driven models. This section reviews efforts to develop nontrivial lumped parameter models. Often, these studies have as a primary focus something other than the two-phase heat exchanger dynamics (e.g., system coefficient of performance [COP], cabin temperature, etc.), and so the model development is treated superficially (e.g., Stoecker 1966; Qi and Deng 2008). However, the first notable effort to develop a nontrivial lumped parameter model that considers multiple fluid regions in the heat exchanger, albeit temporally invariant, was Chi and Didion (1982). However, like most publications in this area, limited experimental validation was presented. Sami et al. (1987) utilized a similar approach, including conservation of momentum in dynamic equations between each control volume and presenting qualitative agreement for a start-up experiment. Ibrahim (2001) used the lumped parameter approach to model a TEV-controlled evaporator and provide simulation results for changes in chilled water inlet temperature. Several additional authors presented models for single-evaporator (Deng and Missenden 1999; Underwood 2001; Browne and Bansal 2002; Chen and Deng 2006) and multiple-evaporator systems (Stack 2002; Wu et al. 2005) based on a lumped capacitance, stirred-tank approach. For those that presented experimental validation, the results showed partial qualitative agreement, but each of the models was clearly unable to capture higher-order dynamic transient behavior in some cases. Llopis et al. (2008) used a lumped parameter approach but allowed the volume of each fluid region (subcooled liquid, two-phase fluid, superheated vapor) to be dynamic; it is unclear whether the heat transfer surface area for each region is also allowed to vary. However, the resulting model was perhaps closer to the MB formulations than the strict lumped parameter approaches.

MB models

In contrast with the FCV, which achieves greater accuracy through finer discretizations, the MB approach seeks to capture the salient dynamics of multiple fluid phase heat exchangers while preserving the simplicity of lumped parameter models. This approach assumes a time-varying boundary between regions of different fluid state (i.e., subcooled liquid, two phase, or superheated vapor). Separate control volumes are considered for each of the fluid regions, and the necessary distributed parameters are lumped for each of these regions (Figure 3).

The focal point of this modeling approach is the ability to predict the effective position of phase change in a heat exchanger. Among the first studies of transient behavior along these lines regarding the transient response of the effective dry-out point in evaporating flows were Wedekind and Stoecker (1966) and Wedekind (1971). Later, using a mean void fraction was proposed to develop a transient model for the evaporating and condensing flows of a VCS (Wedekind et al. 1978; Beck and Wedekind 1981). This research played a critical role in the MB method of modeling vapor compression cycles. The mean void fraction assumption is applied almost universally by other researchers developing MB models, allowing the two-phase region to be modeled in lumped form. A void fraction is defined as the ratio of vapor volume to total volume, and many experimental correlations have been proposed for predicting void fraction for various conditions and fluids; reviews of well-known correlations can be found in Rice (1987) and Wilson et al. (1998). Independent of the correlation used, most authors assume that the value of mean void fraction is time invariant for small transients.

The calculation of mean void fraction is given by

$$\overline{\gamma} = \frac{1}{x_2 - x_1} \int_{x_1}^{x_2} \gamma(x)\, dx. \tag{1}$$

The function $\gamma(x)$ is defined using an assumped correlation. For example, using the well-known "slip-ratio" correlation,

$$\gamma = \frac{x}{x + (1 - x)\mu_s} \tag{2}$$

where $\mu_s = (\rho_g/\rho_f)S$, and S is the slip ratio or the ratio of the vapor velocity to the liquid velocity in a two-phase flow (e.g., assuming homogeneous flow, $S = 1$). If the correlation used is fairly simple, a closed-form solution to the calculation of a mean void fraction can be derived (Rasmussen 2005). For more complex coorelations, a interpolation table of precalculated values (McKinley and Alleyne 2008) or a approximation based on numerical integration may be used (Zhang and Zhang 2006). This assumption has obvious implications for transient prediction of the location of refrigerant charge. The amount of refrigerant charge is highly sensitive to the void fraction (Jensen and Tummescheit 2002), and thus, tuning of the void fraction correlation is

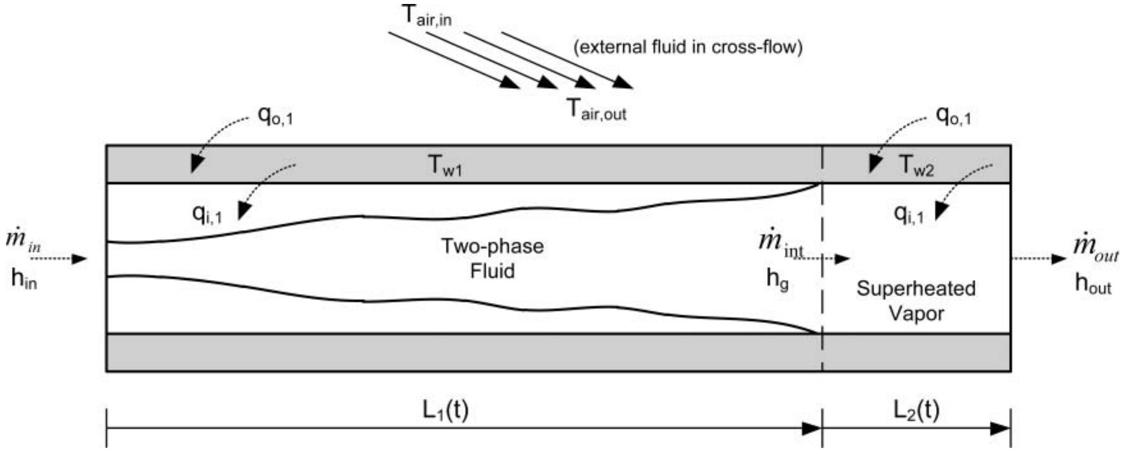

Figure 3. MB evaporator model.

important if quantitative charge prediction is desired (Bendapudi 2005).

Also essential to the MB formulation is the assumption of uniform pressure throughout the heat exchanger. Without exception, all researchers uniformly neglect conservation of momentum, assuming the pressure drop in the heat exchanger to have a negligible effect on the dynamic response and allowing thermodynamic properties to be calculated from the predicted pressure. The only exception is Tian et al. (2005) and Tian and Li (2005), where static pressure drop is included, but the lumped themodynamic properties are still evaluated at isobaric conditions. This would appear to be an adequate solution—including pressure losses between components—but neglecting these effects in the heat exchanger model to minimize the numerical stiffness of the resulting model.

A final item to note when deriving the MB heat exchanger model is the assumption regarding the wall temperature at the interface between two fluid regions. Although the wall temperature for each fluid region is lumped, the transient nature of the boundary between these regions requires that the energy transferred as the regions shrink or expand leads to variations in the assumed interface temperature (Table 1). Most assume that the interface temperature is equal to the two-phase wall temperature, but the most recent publications either switch the assumed interface temperature based on whether the region is growing or shrinking, and still others assume a weighted average. Each of these has their proclaimed advantages, based on either physical interpretation or computational robustness.

Wedekind et al. (1978) presented comparisons of model and experimental data sampled at approximately 4 Hz and showing good agreement between model and data. Unfortunately, in the MB publications since that time, few have presented complete validation results, often presenting only predictions for cooling or operating pressures and omitting comparisons for the superheat or subcool transients that are the focus of the MB approach. Additionally, for some studies, the experimental data is sampled at frequencies so low as to question if significant transient behavior is being missed.

Dhar and Soedel (1979) developed a dynamic model of an entire air-conditioning system using a lumped parameter approach, where mass and energy transfer between the fluid regions is similar to the MB approach, but they presented only simulation results. Yasuda et al. (1983) used a similar approach and presented adequate experimental validation. Broersen and van der Jagt (1980) employed an MB model of an evaporator to analyze hunting of TEVs (no validation). Kapadia and Wolgemuth (1984) presented an MB model for a condenser in the Rankine cycle (limited validation). Grald and MacArthur (1992) followed their work on distributed heat exchanger models with the presentation of an MB model of an evaporator. Limited model validation was presented for an MB evaporator incorporated into an overall heat pump model with a discretized condenser model. Pressure predictions from the original discretized system model was compared to data for compressor startup and shutdown sampled at approximately 0.1 Hz. The discretized system model was then compared to the system model with an MB evaporator for only evaporator cooling capacity and compressor power.

The first to suggest the use of MB models for multivariable control design was He et al. (1997, 1998), who presented linearized MB models for the evaporator and condenser with adequate validation, and reduced-order models were used for multiple inputs-outputs (MIMO) control design. The experimental results demonstrated improved performance for coordinated control strategies.Following these publications, several authors presented virtually identical models (Jensen and Tummescheit 2002; Leducq et al. 2003; Kumar et al. 2008; Lei and Zaheeruddin 2005; Tian et al. 2005; Tian and Li 2005; Rasmussen and Alleyne 2004). Jensen and Tummescheit (2002) modified the model to use a mean liquid fraction instead of a mean void fraction and presented simulation results. Lei and Zaheeruddin (2005) modeled a TEV-controlled chiller system using the MB approach and, likewise, only presented simulation results. Schurt et al. (2009) used an MB model for a double-pipe counter-flow heat exchanger but neglected heat exchanger thermal capacitance favoring a steady-state e-NTU approach instead. Validation evidences that the model matches in steady state, but some transient trends are not emulated, despite the fact that the validation dataset was used to identify more than a dozen model parameters. Tian et al. (2005) and Tian and Li (2005) used an MB model in association with a variable displacement compressor to evaluate stability and sensitivity to compressor parameters; validation shows qualitative, but not quantitative, agreement. MB models were used for a transcritical CO_2 vapor compression cycle with comparison between nonlinear, linearized, and data-driven models

(Rasmussen et al. 2003; Rasmussen and Alleyne 2004; Rasmussen et al. 2005).

Although the MB approach yields a relatively low-order dynamic model (five ordinary differential equations [ODEs] for an evaporator with superheat, seven for a condenser with subcool), many authors have suggested that the dominant dynamics can be described with even fewer equations. Sheng et al. (1996) and Jensen and Tummescheit (2002) utilized an eigenvalue analysis of the linearized model to infer that the two-phase region heat transfer dominates the response. Leducq et al. (2003) similarly assumed that heat transfer in the two-phase region dominates the transient response and proceeds with a model reduction approach similar to Sheng et al. (1996) and includes comparisons with data. Both eigenvalues (Rasmussen and Alleyne 2004) and system identification techniques (Rasmussen et al. 2005) have been used as justification for model reduction, and a transformation of state variables was used to illustrate that the fast dynamics are associated with the conservation of refrigerant energy with comparisons of nonlinear, linear, and reduced-order models to data. Other authors made assumptions regarding the dominant physical process and simplified the models in alternative ways. Kumar et al. (2008) utilized the MB paradigm to model a VCS and an air-handling system, but they neglected convective heat transfer in the heat exchanger vapor region, thus reducing the number of dynamic variables used in the model. Validation is limited to cabin temperature and humidity, and no direct validation of the VCS model is available. In contrast, Eisenhower and Runolfsson (2009) assumed quasi-static flow (inlet and outlet refrigerant flow rates in the heat exchanger are equal) to reduce model order, yielding a single time constant for the evaporator refrigerant dynamics. This is then combined with a time constant for the evaporator metal thermal capacitance. The overall model is of a transcritical system and is used to analyze dynamic bifurcations. The model is calibrated with extensive steady-state and dynamic data and then cross-validated, demonstrating that transients are largely captured, with some steady-state bias errors.

Recent innovations within the MB paradigm have focused on extensions for heat exchangers operating under conditions where the fluid phase regions disappear or reappear, termed switched MB (SMB). Pettit et al. (1998) and Willatzen et al. (1998) handled this by replacing nonactive differential equations with appropriate substitutions, ensuring that numerical difficulties (dividing by zero) are avoided by including small, negligible numerical terms (a similar approach was also used by Bendapudi et al. [2008] but was still unable to handle start-up transients). Gordon et al. (1999) handled the same difficulty instead by actually reducing the number of differential equations in real time, resulting in a set of differential algebraic equations (DAEs). Zhang and Zhang (2006) also presented a switching-based approach, but although the switching conditions are simpler, the paper only includes a mechanism for switching one direction. Except for Bendapudi et al. (2008), none of these publications presented experimental validation of the models, presenting only comparisons against other simulations. McKinley and Alleyne (2008) used a switching strategy for transitioning between MB models as fluid regions disappear or reappear, and subsequent works included extensive validation (Li and Alleyne 2010). An alternative to these models was proposed by Eldredge et al. (2008), where the notion of the mean void fraction is extended to handle small transient deviations from saturation conditions and is most appropriate for heat exchangers with receivers/accumulators. This extended MB (EMB) approach is experimentally validated.

Fixed control volumes models

Finite difference (FD) or FCV approaches to dynamic modeling of two-phase heat exchangers are becoming more prevalent and are available in commercial software packages, e.g., e-Thermal (Anand et al. 2004), Modelica (Tummescheit and Eborn 1998; Tummescheit et al. 2005), SINDA/FLUENT (Cullimore and Hendricks 2001), etc. The attractiveness of this approach is the ability to model the fluid behavior with extreme detail, capturing thermo-physical gradients and distributed parameters. This, reportedly, lends itself to higher accuracy prediction than the lumped parameter approaches that rely on effective parameters and only describe the dominant physical mechanisms for heat transfer, fluid flow, etc. The governing equations are typically derived either by starting with discetizing the heat exchanger into many controls and directly applying conservation equations, assuming average or lumped parameters or by directly discretizing the governing PDEs using the FD method. The end result is similar in form, and the resulting models are both parametrically complex and of high dynamic order.

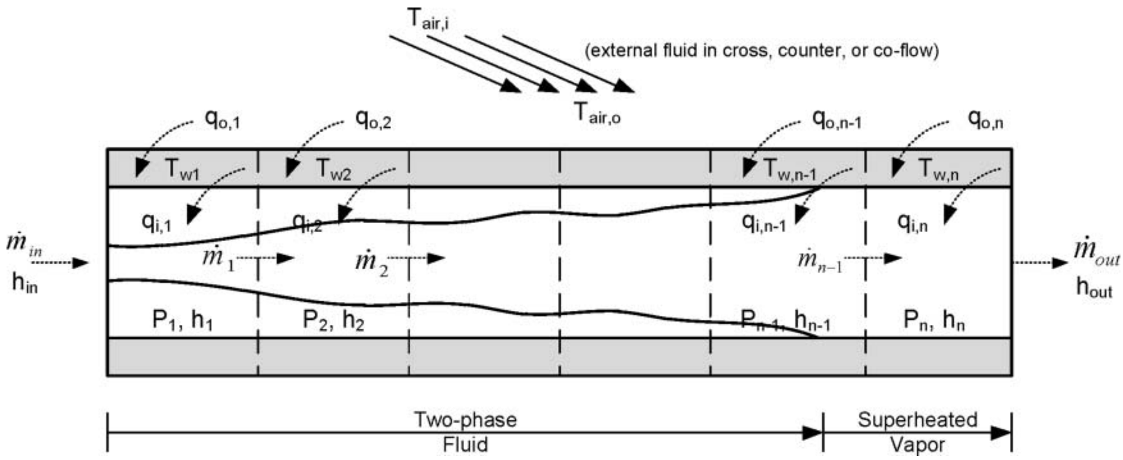

Figure 4. FCV evaporator model.

In contrast to the MB approach, which employs a mean void fraction for predicting the amount of refrigerant mass in the two-phase region, the FCV approach assumes average values within a control region, relying on high levels of discretization to ensure accuracy. Thus, prediction of refrigerant charge in a heat exchanger will depend heavily on the level of discretization and the assumptions employed in calculating the average properties for each region.

Also, unlike the MB publications, where conservation of momentum is uniformly neglected, the FCV/FD publications offer a range of approaches for handling this effect. If a pressure gradient is assumed, then a staggered grid approach for modeling the mass and energy flow for each control region is typical (Eborn 2001), where pressure differentials dictate mass flow rates, and mass flow rate differentials drive the pressure and energy dynamics (Figure 4). The conservation of momentum can be included and incorporate transient effects, resulting in a additional dynamic equations and a numerically stiff system. A second approach is to compute static pressure drops between each region, resulting in a set of DAEs. Finally, the aggregate pressure drop can be computed and included only at the end of the heat exchanger model or simply neglected completely. Table 2 indicates which of the cited publications include or neglect these effects. Zhang et al. (2009) presented a direct comparison of the most common approaches for handling conservation of momentum, revealing that including momentum effects is not critical for large transient simulations but that discernible differences are present for instaneous step changes in mass flow rate. However, as most compressors and valves do not respond instantly, this may not be a critical effect to capture.

The first notable model using this approach was reported by Gruhle and Isermann (1985), which used a discretized model of the evaporator for model-based design of PI valve control strategy; no validation of the model is presented. MacArthur (1984) presented a discretized model and later gave extensive experimental validation of the model (MacArthur and Grald 1989). Wang and Touber (1991) presented a detailed distributed model of a dry expansion evaporator, citing that accurate local modeling of a void fraction and heat transfer is necessary for proper predication. Sami and Comeau (1992) presented a distributed model of a heat pump system, but the validation exhibited some notable discrepancies. The work by Nyers and Stoyan (1994) and Jia et al. (1995, 1999) illustrated one of the possible disadvantages of discretized approaches, namely, that depending on the assumptions employed and how the model is formlated, the resulting set of DAEs may require an iterative solver be used at each time step, significantly increasing the computation time. This may be exacerbated in heat exchangers with counter-flow orientations, where cell-by-cell simultaneous solution is not possible, and a two-level iteration method may be required. However, a particular strength of the discretized approach is its applicability to a variety of flow geometries versus the MB or lumped parameter approaches, where these details are typically lost amidst the simplifying assumptions.

In the last decade, the use of discretized modeling approaches has increased significantly, presumably due to the computational advances and software programs that facilitate the formulation and simulation of the models. Mithraratne et al. (2000) and Mithraratne and Wijeysundera (2001, 2002) presented a discretized model of a TEV-controlled evaporator with validation and a simulation study of TEV-induced hunting. Bendapudi and Braun (2002a) presented a distributed model of a liquid chiller system with validation. Koury et al. (2001) presented simulation results for step changes in compressor speed (without validation). Kim et al. (2004) utilized finite volumes for a heat pump system, but the results revealed the dominance of the hot water reservoir on the overall transient response. García-Valladares et al. (2004) utilized a detailed approach for double-pipe heat exchangers, including a radial discretization of the pipe insulation. Although the temporal response is examined at a few values to determine numerical convergence, only steady-state predictions were compared against data. Kapadia et al. (2009) used discretized models to compare startup reponses of systems with different refrigerants, and Beghi and Cecchinato (2009) used a discretized model for tuning EEV control algorithms. The recent work by Ndiaye and Bernier (2010) sought to model on/off operation of the fans and compressor, including natural convection and humidity effects. This approach has also been applied to transcritical system modeling by Skaugen and Svensson (1998), Pfafferott and Schmitz (2000), and Rigola et al. (2005), although none include transient validation. Tso et al. (2006a, 2006b) presented a validated discretized model for predicting evaporator frosting, and Chen and Lin (1991) and Hermes and Melo (2008) both applied this method to model small refrigeration systems. Discretized dynamic models are also commonly used for heat exchangers in other HVAC-related systems, such as absorption systems (Kohlenbach and Ziegler 2008a, 2008b; Matsushima et al. 2010), adsorbption systems (Schicktanz and Núñez 2009), chilled water coils (Zhou and Braun 2007a, 2007b; Li et al. 2010), among others.

Beyond the varied systems where discretized models have been applied, a few notable publications presented some modeling innovations. Bendapudi et al. (2008) provided a comparison of MB and discretized approaches, concluding that the finite-volume approach was more robust, the MB approach was faster computationally, and that the two paradigms generally resulted in equivalent responses. Wang et al. (2007) presented a discretized model that separately addressed the gas and liquid phases in the two-phase region of the heat exchangers in lieu of the more common void fraction approach and a custom iterative solution method to reduce computation time. Unfortunately, the validation shows that while gross pressure trends are captured, the temperatures have notable errors (the authors cite incorrect parameter and boundary conditions) and miss significant transients.

A recent innovation with particular impact on FCV model implementation is the use of object-oriented models. The evolution of simulation software significantly influences the form and substance of modeling efforts. The shift from line-by-line code (e.g., Fortran routines) to block diagram-based modeling tools (e.g. Simulink©) helped shape the transition from system-level modeling efforts to modular component-focused simulations with a signals-and-systems perspective. In this framework, defining causality, interconnection variables, and avoiding algebraic loops (i.e., circular algebraic dependencies) becomes critical. The most recent trend in physics-based modeling is object-oriented modeling, where component interconnections represent actual physical connections. This trend is clearly reflected in recent publications on vapor compression simulation (Tummescheit et al. 2000, 2005; Jensen and Tummescheit 2002; Pfafferott and Schmitz 2004; Schicktanz and Núñez 2009; Zhang et al. 2009; Li et al. 2010; Matsushima et al. 2010). The prominent object-oriented software tools (e.g. Modelica, Simscape) require the user to define component governing equations but not causality relationships between variables or systems. Issues of causality and intercomponent dependencies are determined by the simulation software. Additionally, the simulation software typically solves the component initial conditions implicitly using system-level input values rather than requiring the user to specify component initial conditions and ensuring matching boundary conditions. Finally, these software packages specialize in solving DAEs that arise naturally in physics-based modeling, alleviating the need for the user to analytically eliminate algebraic relationships and multi-time scale dynamics through model reduction or approximate simulation techniques. Although improved DAE algorithms are a continuing priority in the simulation community, current solvers are reasonably robust and computationally efficient for the typical VCS simulation.

Before concluding this section on discretized approaches, a reference to bond graph approaches is appropriate. Classical bond graph models utilize power and energy to graphically describe system dynamics in multiple domains (Karnopp et al. 1990; Brown 2006). However, thermo-fluid energy systems pose a particular challenge, leading to the use of pseudo-bond graphs for compressible fluid systems (Karnopp 1979), where standard energy flows are replaced by thermal/enthalpy/hydraulic flow variables, and energy states are replaced with pressure/temperature/etc. states. To deal with multiphase flows, additional variables are needed. Although the pseudo-bond graph method is capable of capturing the dynamics of these systems, the authors describe the process of selecting state variables as "not trivial" (Bouamama 2003) and the process of defining connections as "awkward" (Brown 2006). Despite these challenges, bond graph models for the VCS were developed by Shoureshi and McLaughlin (1984) and included some nominal experimental validation. More recently, Lin and Yeh (2007b) utilized a bond graph framework with experimentally identified parameters. But these efforts appear mostly isolated from more conventional approaches.

Real-time simulation

Modern computing technology is rapidly advancing the capability for real-time simulation of complex dynamical systems. However, even with current capabilities, real-time simulation may not be possible depending on the modeling paradigm, time scale, and numerical integration algorithm selected. Only a portion of the publications cited contain details regarding the numerical simulation of the proposed models. These details are summarized in Table 3. The real-time factor is computed by dividing the execution time by the simulated time. As is clear from the reported results, the lumped parameter and MB approaches generally result in simulations that run faster than real time, whereas the FCV models typically are much slower than real time. This is naturally dependent on the level of discretization and whether multi-time scale behavior is included (e.g., conservation of momentum), necessitating small time steps, variable step solvers, or DAE solvers. Part II of this article provides a detailed comparsion of computation times for MB and FCV models for identical systems.

Data-based dynamic models

In contrast with physics-based modeling approaches, the primary advantages of data-driven dynamic models are typically expediency and accuracy. Identifying a simple dynamic model from data is generally faster than developing, tuning, and validating a physics-based model. Moreover, as the identified model is created directly from experimental data, the models are expected to be quantitatively, not merely qualitatively, accurate. However, this approach is not without its limitations. Chief among these is a lack of robustness; identified models are typically only valid within the prescribed operating range represented by the data and only for the particular system used. Changes in operating conditions or components may invalidate the model, and the process must be repeated.

Additionally are the challenges inherent to any data-driven modeling task. Only a few are mentioned here, and the interested reader is referred to Ljung (1999) for a more detailed treatment of these issues. The first consideration is selection of the model structure and dynamic order. This choice should be guided by physical insight to the process, and the choice should be parsimonious to avoid "over-fitting" of the model to the data. Although there is always a tradeoff between model accuracy and model order, the simplest (lowest-order) model that achieves the required accuracy should be selected. The second challenge is ensuring that the transient experiment is designed appropriately. The choice of input excitation signals should be chosen with appropriate magnitude (larger magnitudes generally accentuate nonlinear behavior, smaller magnitudes result in smaller signal-to-noise ratios) and frequency content (input signals should excite the desired dynamics). Care must be taken when the system is identified while under closed loop control, as this poses a particular challenge for data-driven modeling (Ljung 1999).

Although data-driven approaches are most commonly applied when a single input–output relationship is needed, occasionally empirical models are included as part of a larger system model, resulting in a gray-box model. Estrada-Flores et al. (2006) provides a good review of techniques and challenges associated with gray-box methods, including integration strategies and validation metrics.

As applied to VCSs, the types of data-driven modeling approaches in the literature include linear models, such as transfer functions, time-series models (ARX, ARMAX, etc.), and state-space models, and nonlinear models such as linear parameter varying (LPV) models, gain-scheduled transfer functions, and Wiener/Hammerstein models. Use of more complex nonlinear methods, such as fuzzy

Table 3. Summary of models with numerical simulation results

		Heat exchanger			Computation		
Citation	System	Paradigm	Momentum	Heat exchanger discretization	Solver	Step size	Real-time factor
Beghi and Cecchinato (2009)	Fin-coil evaporator	FD	Included	NA	Semi-implicit Euler (stiff solver)	NA	NA
Bendapudi et al. (2008) Bendapudi (2005)	Centrifugal chiller	FCV	Neglected	1–40 cells (15 rec.)	First-order Euler Second-order Euler (rec.) Fourth-order Runge-Kutta	0.0025–0.080 s (0.04 s rec. for condenser) (0.01 s rec. for evaporator)	1.2
		SMB	Neglected			0.01–0.04 s (0.02 s rec.)	0.35–0.55
Browne and Bansal (2001)	Liquid chiller	Lumped	Neglected		Fifth-order Runge-Kutta	Variable step	NA
Chi and Didion (1982)	Residential heat pump	Lumped	Neglected		Euler	0.005 s	NA
Dhar and Soedel (1979)	Refrigeration system	Lumped	Neglected		Euler	0.001 min	NA
Estrada-Flores et al. (2003)	Receiver/ accumulator				Fourth-order Runge-Kutta	0.1 s	NA
García-Valladares et al. (2004)	Double-pipe HX	FCV	Included	10–2000 cells (100–200 rec.)	Iterative solver	0.25–200 s (10 s rec.)	NA
Hermes and Melo (2008)	Household refrigerator	FCV	Neglected	64 cells (evaporator) 76 cells (condenser)	Adams	Variable step	NA
Jia et al. (1995) Jia et al. (1999)	Evaporator	FCV	Included	180 cells	Newton-Raphson	5 s	2.4
Koury et al. (2001)	Refrigeration system	FD	Included	400 cells	Fourth-order Runge-Kutta	15 s	NA

Kumar et al. (2008)	Railway soach air conditioner	MB	Neglected		Fourth-order Runge-Kutta	NA	NA
Leducq et al. (2003)	Refrigeration system	MB	Neglected		Second-order Newton-Raphson	2 s	0.042
Llopis et al. (2008)	Shell-and-tube condenser	Lumped	Neglected		Fourth-order Runge-Kutta	0.2 s	0.018
Mithraratne et al. (2000) Mithraratne and Wijeysundera (2001) Mithraratne and Wijeysundera (2002)	Counterflow evaporator	FD	Neglected	NA	Implicit backwards difference	NA	NA
Tso et al. (2006a) Tso et al. (2006b)	Evaporator with frosting	FD	Included	240 cells	Newton-Raphson	NA	NA
Tummescheit et al. (2005)	Vapor compression	FCV	Included	User specified	DASSL DAE solver	NA	NA
Wang et al. (2007)	Refrigeration system	FD	Neglected	NA	Triangle and tri-point	NA	NA
Wang and Touber (1991)	Plate fin air cooler	FD	Included	60 cells	NA	0.1 s	10

rec. = recommended

models and neural networks, is extremely limited at this point and is more common for modeling simpler HVAC&R components, such as air-handling units or variable air volume systems.

Modeling paradigms

Linear models

A common approach for modeling of VCSs where only one input–output is considered (e.g., EEV to evaporator superheat) is a process control model. The basic form of this model includes a gain $k_{\mathbf{0}}$, a first-order dynamic characterized by the time constant τ, and a pure time delay T_d. This basic model is easily augmented with additional time constants as needed:

$$G(s) = \frac{k_{\mathbf{0}} e^{-T_d S}}{\tau s + \mathbf{1}}. \tag{3}$$

Stoecker (1966) utilized this approach when characterizing the dynamics of a TEV-driven evaporator, where the evaporator is modeled with two first-order dynamics with a time delay and an additional dynamic for the valve. The associated parameters were identified for various operating conditions using experimental data. Yasuda et al. (1993) used a process control model to represent the relationship between compressor speed and cooling capacity. Outtagarts et al. (1997) assumed a classical process control model for an EEV-evaporator dynamics, but they then approximated the time delay as a first-order dynamic for simplicity in analysis and design. Again the parameters dependent on the operating condition were characterized by curve fits to experimentally identified values. Finn and Doyle (2000) identified two different process control models for the same EEV-evaporator loop at high and low cooling loads. Mulay et al. (2005) analytically derived an approximate time constant associated with the sensing bulb of a TEV, and they then explored the impact of bulb location and time constant on superheat control. More recently, Fallahsohi et al. (2010) utilized a physical model to identify the approximate parameters of a process control model for use with predictive control strategies. Elliott and Rasmussen (2010) utilized second-order models with varying parameters to capture EEV-superheat and EEV-pressure dynamics. Broersen and ten Napel (1983) presented an identified model of the TEV's sensing bulb dynamics using frequency domain data, resulting in a second-order dynamic model, later used in dynamic analysis for stability (Broersen 1982). Lin and Yeh (2007a) used a pseudo-bond graph to motivate the assumption of an experimentally identified first-order model of a VCC with both valve and compressor inputs, and superheat and evaporation temperature outputs. Following the same procedure, a subsequent paper (Lin and Yeh 2007b) extended this approach to a multi-evaporator system, but the validation plots are less convincing.

More classical identification techniques have also been used to model VCC dynamics using linear discrete time models (Equation 4), where the filter coefficients are determined from data, i.e., techniques presented in Ljung (1999):

$$\begin{aligned} y(t) &+ a_{\mathbf{1}} y(t-\mathbf{1}) + \cdots + a_{n_a} y(t-n_a) \\ &= b_{\mathbf{1}} u(t-\mathbf{1}) + \cdots + b_{n_b} u(t-n_b) + c_{\mathbf{1}} e(t-\mathbf{1}) \\ &\quad + \cdots + c_{n_c} e(t-n_c). \end{aligned} \tag{4}$$

Bechtler et al. (2001) utilized a nonlinear autoregressive moving average (ARMA)-based model (i.e., $e(t) = \mathbf{0}$, no external noise signal is considered) with the nonlinear functions identified using an artificial neural network. Validation is good for some variables, with some notable discrepancies for others. In a subsequent paper, Browne and Bansal (2001) compared this model to a lumped parameter physics model, with simple modes based on the thermal capacitance of refrigerant, heat exchanger metal, and cooling loads. The comparative validation of the models is not uniform, with the physics model appearing to demonstrate better accuracy than the neural network based model. Navarro-Esbrí et al. (2006) utilized a strictly regressive model (i.e., $a_i = 0, \forall i$, Equation 4), with parameter values that are updated in real time and estimated using the covariance matrix and a forgetting factor identification approach. The objective is a model of VCS chiller dynamics for the purpose of fault detection. More advanced system identification techniques have also been applied, such as in Shah et al. (2004), where a canonical state space representation is used to model an MIMO VCS.

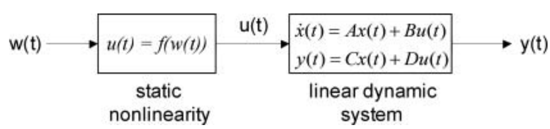

Figure 5. Hammerstein model.

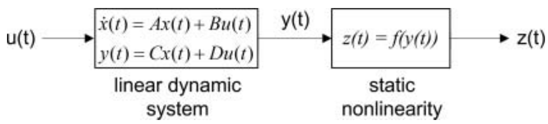

Figure 6. Wiener model.

Nonlinear models

As with most HVAC&R systems, VCSs exhibit nonlinear dynamics. Although these can be approximated with linear models, they are typically only valid for a small operating range. To extend model validity, a few authors have reported nonlinear data driven models. Two common approaches include Hammerstein/Wiener models (i.e., linear identified models with a nonlinear input or output function, Figures 5 and 6) and LPV models, where parameter values are determined at multiple conditions and then scheduled based on a measured signal (e.g., Equation 5). Examples of these approaches include Outtagarts et al. (1997) and Finn and Doyle (2000), where parameters defining the relationship between expansion valve and evaporator superheat and pressure are determined at multiple operating points and then interpolated:

$$G(s, \theta) = \frac{k(\theta)}{\tau(\theta)s + \mathbf{1}}. \qquad (5)$$

Other approaches, such as fuzzy models or neural networks, appear in the literature but are primarily used for control rather than dynamic modeling. For example, Nanayakkara et al. (2002) presented a neural network approach for controlling an ammonia-based evaporator, but the model is a two-region (fixed) lumped parameter model with a sub-neural network trained to produce one of the model outputs.

Conclusion

The literature regarding VCS dynamics is extensive, with a wide array of approaches proposed. Users should carefully select the appropriate modeling paradigm based on the intended use of the model. Because of the uncertain nature of some VCS model parameters, experimental validation is integral to every modeling effort. While existing models can capture salient VCS dynamics in the majority of scenarios, continuing efforts in the areas of model reduction, numerical solvers, and automated model tuning and validation are necessary to increase the capabilities of these models for analysis, design, control design, and fault detection.

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